PRN No.	Total No. of Questions: 09

QP Code:

JSPM's

Rajarshi Shahu College of Engineering, Tathawade, Pune- 411033

(An autonomous institute affiliated to Savitribai Phule Pune University)

END SEMESTER EXAMINATION (ESE), DEC./JAN. -2023

Academic Year: 2023-24(SEM-I)

Class: F.Y. B. Tech.(Civil, Mechanical, E&TC, Electrical, A&R)

Subject Name and pattern: ENGINEERING MATHEMATICS-I [ES1201]

Duration: Hours **Max. Marks: 50 Marks**

Instructions to the Candidates

1. **Q.1, Q. 2 and Q.3** is compulsory.
2. Attempt **Q.4 or Q.5 , Q.6 or Q.7 , Q.8 or Q.9.**
3. Figures to the right indicates full marks.
4. Use of nonprogrammable electron pocket calculator is allowed.
5. Assume suitable and necessary data wherever required.

Q. No.			Marks	BL	CO
1		**Attempt (Any One)**			
	a	Find the value of k so that the equations $x+y+2z=6$; $x+2y+4z=10$; $x+4y+8z=k$ have infinite number of solutions. Hence find the solution.	4	BL3	CO1
	b	Determine the values of a, b, c when $\begin{bmatrix} 0 & 2b & c \\ a & b & -c \\ a & -b & c \end{bmatrix}$ is orthogonal	4	BL3	CO1
2		**Attempt (Any One)**			
	a	Find the Eigen values and **Eigen vector corresponding to smallest eigen value** of the matrix $A=\begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$	4	BL3	CO1
	b	Find the modal matrix P which diagonalize the matrix $A=\begin{bmatrix} 14 & -10 \\ 5 & -1 \end{bmatrix}$	4	BL3	CO1
3		**Attempt (Any One)**			
	a	Using Taylors theorem, expand $2x^3+7x^2+x-6$ in power of $(x-3)$	4	BL3	CO2
	b	Evaluate : $\lim_{x\to\frac{\pi}{2}}(\cos x)^{\cos x}$	4	BL3	CO2

4 a Express $f(x)=(\pi-x)$ in Fourier series with period 2π defined in $(0,2\pi)$ — 4 BL3 CO3

b Find the Fourier series to represent the function $f(x)=\pi^2-x^2$ in $-\pi<x<\pi$ and $f(x+2\pi)=f(x)$ — 4 BL3 CO3

c The turning moment T units of the crank shaft of a steam engine is given for a series of values of the crank angle θ in degrees. — 4 BL3 CO3

θ	0	30	60	90	120	150
T	0	5224	8097	7850	5499	2626

Find the first two harmonics b_1 and b_2 in a series of sines to represent T. (Take $L=\pi$)

OR

5 a Express y as a Fourier series upto first harmonic where y is given as — 4 BL3 CO3

x	0	$\frac{\pi}{6}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{5\pi}{6}$
y	0	9.2	14.4	17.8	17.3	11.7

b Find the constant term a_0 and coefficient of first two terms a_1 and a_2 of series of cosines to represent y where y is given in the following table (Take $L=6$) — 4 BL3 CO3

x	0	1	2	3	4	5
y	4	8	15	7	6	2

c Obtain the half range sine series expansion of $f(x)=e^x$ in $0<x<1$ — 4 BL3 CO3

6 a If $u=\dfrac{e^{x+y+z}}{e^x+e^y+e^z}$ then find the value of $\dfrac{\partial u}{\partial x}+\dfrac{\partial u}{\partial y}+\dfrac{\partial u}{\partial z}$ — 4 BL3 CO4

b If $u=\sin^{-1}\left(\sqrt{x^2+y^2}\right)$, then prove that — 5 BL3 CO4

$$x^2\frac{\partial^2 u}{\partial x^2}+2xy\frac{\partial^2 u}{\partial x\partial y}+y^2\frac{\partial^2 u}{\partial y^2}=\tan^3 u$$

c If $z=f(u,v)$; $u=\log(x^2+y^2)$; $v=\dfrac{y}{x}$, then show that — 4 BL3 CO4

$$x\frac{\partial z}{\partial y}-y\frac{\partial z}{\partial x}=(1+v^2)\frac{\partial z}{\partial v}$$

OR

7 a If $x^2 = au + bv$; $y^2 = au - bv$; then find the value of 4 BL3 CO4

$$\left(\frac{\partial u}{\partial x}\right)_y \cdot \left(\frac{\partial x}{\partial u}\right)_v + \left(\frac{\partial v}{\partial y}\right)_x \cdot \left(\frac{\partial y}{\partial v}\right)_u$$

b If $u = \operatorname{cosec}^{-1} \cdot \sqrt{\dfrac{x^{1/2} + y^{1/2}}{x^{1/3} + y^{1/3}}}$, then show that 5 BL3 CO4

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = \frac{1}{144} \tan u \left[13 + \tan^2 u\right]$$

c If $u = x^2 + y^2$; $v = 2xy$ and $z = f(u,v)$, 4 BL3 CO4

then show that $x \dfrac{\partial z}{\partial x} - y \dfrac{\partial z}{\partial y} = 2\left(\sqrt{u^2 - v^2}\right) \dfrac{\partial z}{\partial v}$

8 a If $ux = yz$, $vy = zx$, $wz = xy$ then find $\dfrac{\partial(u,v,w)}{\partial(x,y,z)}$ 4 BL3 CO5

b Examine whether the functions $u = \dfrac{x+y}{1-xy}$, $v = \tan^{-1} x + \tan^{-1} y$ are functionally dependent. If dependent then find the relation between them. 5 BL3 CO5

c Discuss maxima and minima for $x^3 + y^3 - 3axy$; $a > 0$. 4 BL3 CO5

OR

9 a If $x = uv, y = \dfrac{u}{v}$ then prove that $J \cdot J' = 1$ 4 BL3 CO5

b Examine whether the functions $u = \dfrac{x-y}{x+y}$, $v = \dfrac{x+y}{x}$ are functionally dependent. If dependent then find the relation between them. 5 BL3 CO5

c The focal length of a mirror is calculated by using $\dfrac{2}{f} = \dfrac{1}{v} - \dfrac{1}{u}$. Find the percentage error in 'f'.if u *and* v are both of error 2 % each. 4 BL3 CO5

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